Ques - Given three integer n, a and b; return n^{th} magical number. Since the answer may be very large return mod $10^9 + 7$.

Soln: Magical no → If it is divisible by a or b.

Test case → $n=1$, $a=2$, $b=3$
out $= 2$

Approach : We have to find a number such that it is divisible by a or b ~~but not~~ or by both.

So, we can use inclusion-exclusion principle here.
if $a = 4$, $b = 6$, $n = 2$
(Let $x = 20$)
Divisible by 4 → $20/4 = 5$
Divisible by 6 → $20/6 = 3$

But numbers divisible by both get counted twice (divisible by LCM)
LCM$(4, 6) = 12$
Divisible by 12 → $20/12 = 1$

Count(x) $= 5 + 3 - 1 = 7$

~~Search~~ Since count$(x) = 7 \geq 2$ ⇒ So valid

Since we have to do searching and Binary search can be used.
Search for count$(x) \geq n$.

Search space → smallest possible magical no, = 1
Largest " " " = n * min(a, b)

Binary search logic
```
mid = (low + high)/2;
if (count(mid) < n) {
    move right (low = mid + 1)
else
    mid could be ans (high = mid)
```
Keep shrinking until low == high.

Low will point to exact n^{th} magical number.

Code:
```
int nthMagicalNumber (int n, int a, int b) {
    Const int MOD = 1e9 + 7;
    long long low = 1
    long long high = (long long)n * min(a, b);
    long long lcm = (long long) a * b / __gcd(a, b);

    while (low < high) {
        long long mid = low + (high - low)/2;
        long long count = mid/a + mid/b - mid/lcm;

        if (count < n) {
            low = mid + 1;
        }
        else high = mid;
    }

    return low % MOD;
}
```

Dry Run

$a = 4$

$b = 6$

$n = 5$

Magical nos → 4, 6, 8, 12, 16, 18 ...

$\gcd(4,6) = 2$

$Lcm = (4 \times 6)/2 = 12$

Search Range

$low = \min(4, 6) = 4$

$high = n \times \min(a,b) = 5 \times 4 = 20$

BS (Iterations)

1. $l = 4$

   $h = 20$

   $mid = (4+20)/2 = 12$

   count magical nos. ≤ 12, count = 4

   we need $n=5$, $5 > 4 \Rightarrow$ move right

   $low = mid + 1 = 13$

2. $l = 13, h = 20, mid = 16$

   count = 5

   $\boxed{5 = n}$ ✓ Enough magical nos.

Try smaller

3. $l = 13, h = 16, mid = 14$

   count = 4

   $4 < 5$, move right

4. $l = 15, h = 16, mid = 15$

   count = 4

   move right

   $l = 16$

   $l = 16, h = 16 \Rightarrow \boxed{answer = 16}$